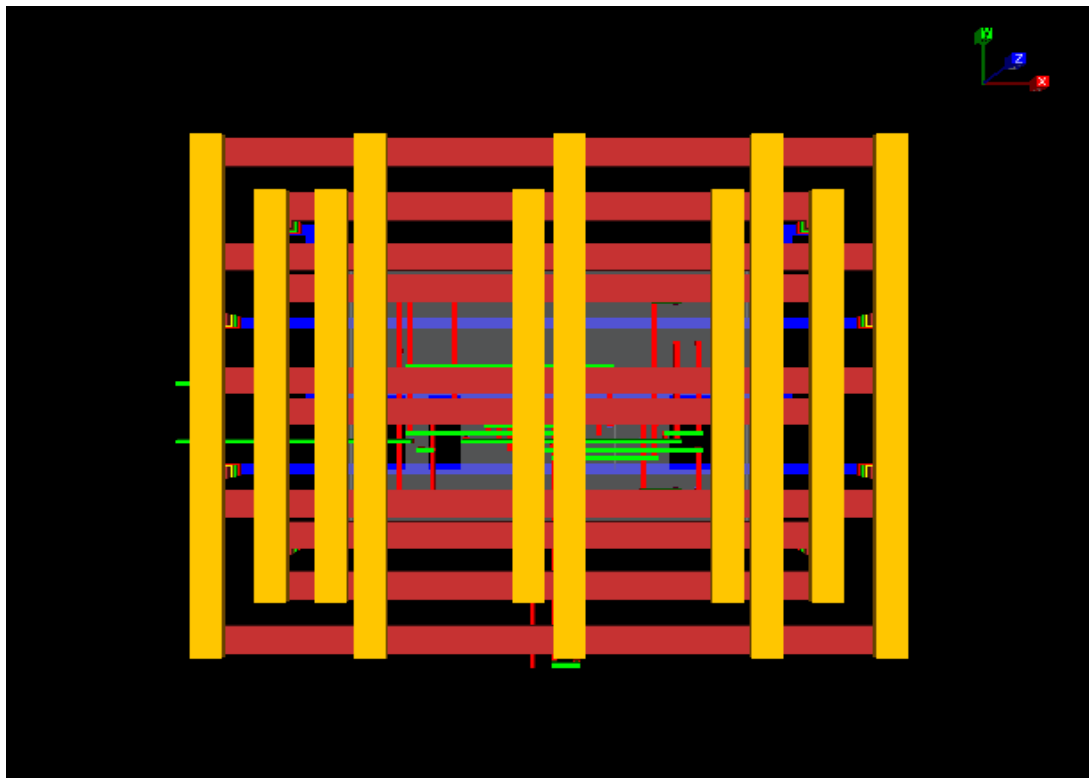
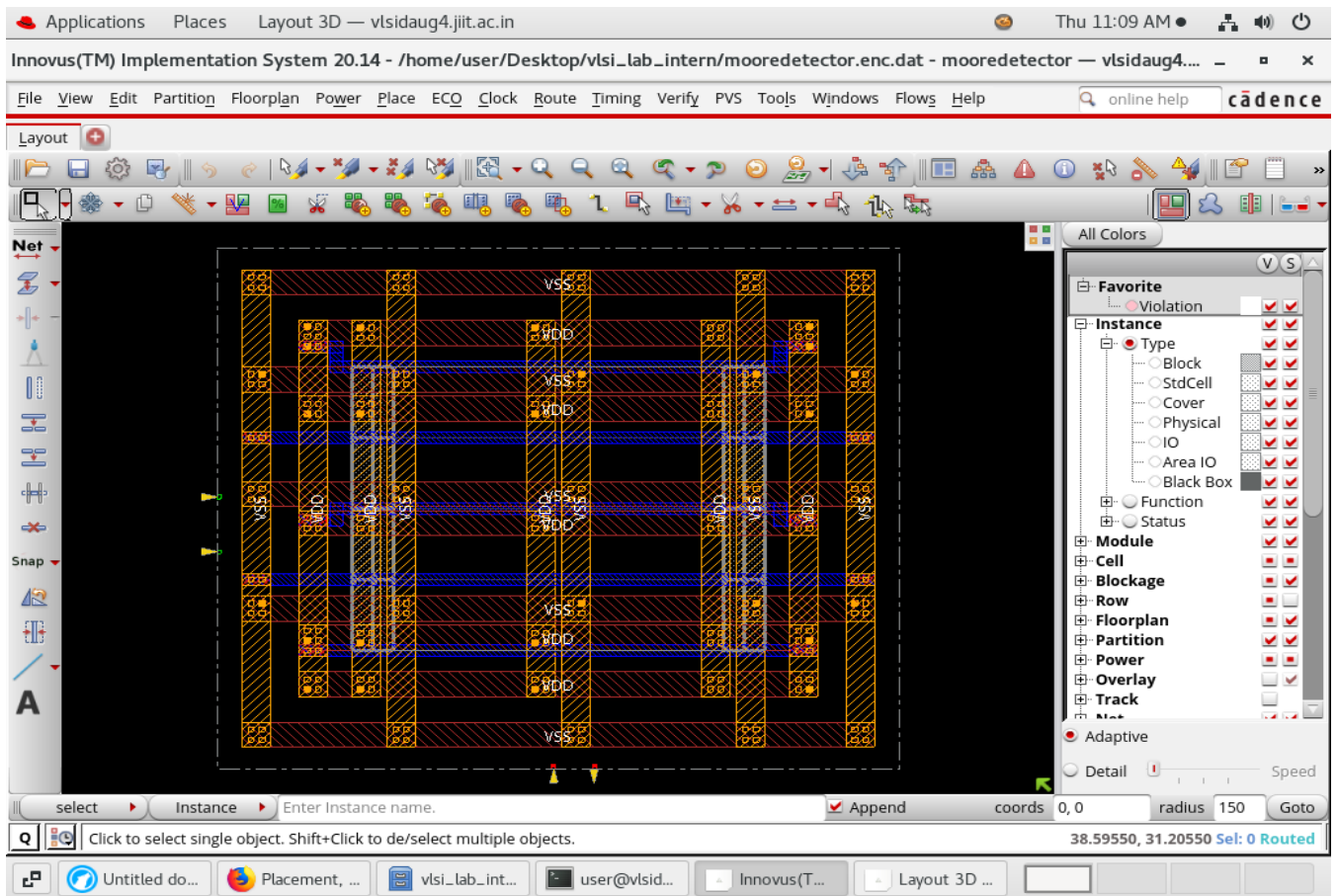


1. FSM - MOORE SEQUENCE DETECTOR



TIMING REPORT

Calculate delays in BcWc mode...

Start delay calculation (fullDC) (1 T). (MEM=1548.07)

Total number of fetched objects 19

AAE_INFO: Total number of nets for which stage creation was skipped for all views 0

End delay calculation. (MEM=1630.96 CPU=0:00:00.0 REAL=0:00:00.0)

End delay calculation (fullDC). (MEM=1603.89 CPU=0:00:00.0 REAL=0:00:00.0)

*** Done Building Timing Graph (cpu=0:00:00.1 real=0:00:00.0 totSessionCpu=0:02:40 mem=1603.9M)

timeDesign Summary

Setup views included:

WORST

Setup mode	all	reg2reg	default
WNS (ns):	0.000	N/A	0.000
TNS (ns):	0.000	N/A	0.000
Violating Paths:	0	N/A	0
All Paths:	0	N/A	0

	Real	Total
DRVs		
	Nr nets(terms)	Worst Vio Nr nets(terms)

+-----+-----+-----+-----+							
max_cap		0 (0)		0.000		0 (0)	
max_tran		0 (0)		0.000		0 (0)	
max_fanout		0 (0)		0		0 (0)	
max_length		0 (0)		0		0 (0)	
+-----+-----+-----+-----+							

Density: 86.719%

Routing Overflow: 0.00% H and 0.00% V

Reported timing to dir timingReports

Total CPU time: 0.1 sec

Total Real time: 0.0 sec

Total Memory Usage: 1571.148438 Mbytes

*** timeDesign #1 [finish] : cpu/real = 0:00:00.1/0:00:00.6 (0.2), totSession cpu/real = 0:02:39.6/0:25:13.0 (0.1), mem = 1571.1M

AREA REPORT

nnovus 1> report_area

Hinst Name	Module Name	Inst Count	Total Area

mooredetector		13	379.490

Power report

Power Net Detected:

Voltage	Name
---------	------

0V	VSS
----	-----

0.9V	VDD
------	-----

Using Power View: WORST.

Set Default Frequency 100MHz.

Begin Power Analysis

0V VSS

0.9V VDD

Begin Processing Timing Library for Power Calculation

Begin Processing Timing Library for Power Calculation

Begin Processing Power Net/Grid for Power Calculation

Ended Processing Power Net/Grid for Power Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1193.47MB/2728.68MB/1429.41MB)

Begin Processing Timing Window Data for Power Calculation

Ended Processing Timing Window Data for Power Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1193.59MB/2728.68MB/1429.41MB)

Begin Processing User Attributes

Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=1193.65MB/2728.68MB/1429.41MB)

Begin Processing Signal Activity

Starting Levelizing

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 10%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 20%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 30%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 40%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 50%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 60%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 70%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 80%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 90%

Finished Levelizing

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

Starting Activity Propagation

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

** INFO: (VOLTUS_POWER-1356): No default input activity has been set. Defaulting to 0.2.

Use 'set_default_switching_activity -input_activity' command to change the default activity value.

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 10%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 20%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 30%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 40%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 50%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 60%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 70%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 80%

Finished Activity Propagation

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

Ended Processing Signal Activity: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1193.91MB/2728.68MB/1429.41MB)

Begin Power Computation

of cell(s) missing both power/leakage table: 0

of cell(s) missing power table: 0

of cell(s) missing leakage table: 0

of MSMV cell(s) missing power_level: 0

Starting Calculating power

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

... Calculating switching power

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 10%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 20%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 30%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 40%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 50%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 60%

... Calculating internal and leakage power

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 70%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 80%

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT): 90%

Finished Calculating power

2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)

Ended	Power	Computation:	(cpu=0:00:00,	real=0:00:00,
mem(process/total/peak)=1194.35MB/2728.68MB/1429.41MB)				

Begin Processing User Attributes

Ended	Processing	User	Attributes:	(cpu=0:00:00,	real=0:00:00,
mem(process/total/peak)=1194.35MB/2728.68MB/1429.41MB)					

Ended	Power	Analysis:	(cpu=0:00:00,	real=0:00:00,
mem(process/total/peak)=1194.43MB/2728.68MB/1429.41MB)				

Begin Boundary Leakage Calculation

Ended Boundary Leakage Calculation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1194.43MB/2728.68MB/1429.41MB)

Begin Static Power Report Generation

```

*-----
* Innovus 20.14-s095_1 (64bit) 04/19/2021 14:41 (Linux 2.6.32-431.11.2.el6.x86_64)
* Date & Time: 2025-Jun-26 11:33:46 (2025-Jun-26 06:03:46 GMT)
*-----
*
* Design: mooredetector
* Liberty Libraries used:
*
*-----
*
*                               WORST:
/home/user/Desktop/vlsi_lab_intern/mooredetector.enc.dat/libs/lib/typ/slow.lib
*
* Parasitic Files used:
*   Power View : WORST
*   User-Defined Activity : N.A.
*   Activity File: N.A.
*   Hierarchical Global Activity: N.A.
*   Global Activity: N.A.
*   Sequential Element Activity: N.A.
*   Primary Input Activity: 0.200000
*   Default icg ratio: N.A.
*   Global Comb ClockGate Ratio: N.A.
*   Power Units = 1mW
*   Time Units = 1e-09 secs
*   report_power
*-----
*
Total Power
*-----
Total Internal Power:    0.00085341    78.5743%
Total Switching Power:   0.00023040    21.2134%
Total Leakage Power:     0.00000231    0.2123%
Total Power:            0.00108612

```

Group	Internal	Switching	Leakage	Total	Percentage
Power	Power	Power	Power (%)		
Sequential	0.0006012	5.512e-05	1.683e-06	0.000658	60.58
Macro	0	0	0	0	
IO	0	0	0	0	
Combinational	0.0002522	0.0001753	6.223e-07	0.0004281	39.42
Clock (Combinational)		0	0	0	0
Clock (Sequential)		0	0	0	0
Total	0.0008534	0.0002304	2.306e-06	0.001086	100

Rail	Voltage	Internal	Switching	Leakage	Total	Percentage
Power	Power	Power	Power (%)			
VDD	0.9	0.0006767	0.0001816	1.935e-06	0.0008602	79.2
Default	0.9	0.0001767	0	3.706e-07	0.0001771	16.3

* Power Distribution Summary:

* Highest Average Power: state_reg[0] (DFFRX4): 0.0002583

* Highest Leakage Power: state_reg[1] (DFFRX4): 5.611e-07

* Total Cap: 4.49202e-14 F

* Total instances in design: 13

* Total instances in design with no power: 0

* Total instances in design with no activity: 0

* Total Fillers and Decap: 0

Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00,

mem(process/total/peak)=1194.70MB/2728.68MB/1429.41MB)